ELECTROMAGNETIC FILEDS AND WAVES

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C02-7

EXPERIMENT 7

AIM:

To determine and plot the capacitance of an isolated sphere and a concentric spherical capacitor for different radii and dielectric constants using MATLAB.

Test Case (i) – Capacitance of an Isolated Sphere, Determine the capacitance for a specified radius and relative permittivity.

Program:

```
clc; clear;

epsilon0 = 8.854e-12;

a = input('Enter radius of isolated sphere (m): ');

er = input('Enter relative permittivity: ');

epsilon = epsilon0*er;

C = 4*pi*epsilon*a;

disp('Capacitance of isolated sphere (F) =');

disp(C);
```

OUTPUT:

```
Enter relative permittivity: 1
Capacitance of isolated sphere (F) =
   5.5631e-11

>>
```

Test Case (ii) – Variation of Isolated-Sphere Capacitance with Radius

PROGRAM:

```
clc; clear;

epsilon0 = 8.854e-12; er = 1;

a = linspace(0.1,2,100);

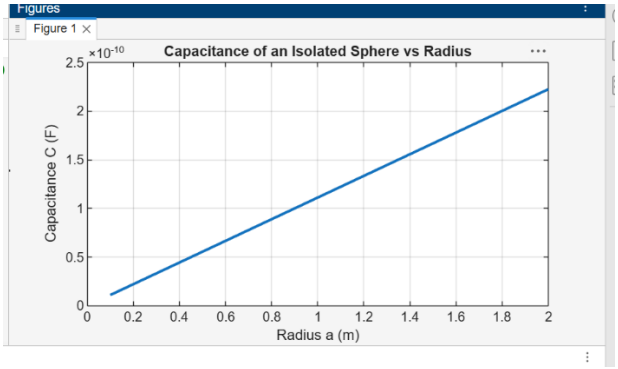
C = 4*pi*epsilon0*er.*a;

plot(a,C,'LineWidth',2); grid on;

xlabel('Radius a (m)'); ylabel('Capacitance C (F)');
```

title('Capacitance of an Isolated Sphere vs Radius');

OUTPUT:



Test Case (iii) – Capacitance of a Concentric Spherical Capacitor, Determine the capacitance for specified inner radius a, outer radius b and relative permittivity εr.

PROGRAM:

clc; clear;

epsilon0 = 8.854e-12;

a = input('Enter inner radius a (m): ');

b = input('Enter outer radius b (m): ');

er = input('Enter relative permittivity: ');

epsilon = epsilon0*er;

C = 4*pi*epsilon*a*b/(b-a);

disp('Capacitance of concentric spherical capacitor (F) =');

disp(C);

OUTPUT:

```
Enter relative permittivity: 2.5
Capacitance of concentric spherical capacitor (F) =
   2.7816e-11

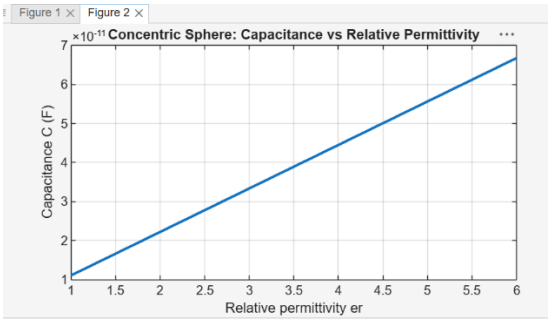
>>
```

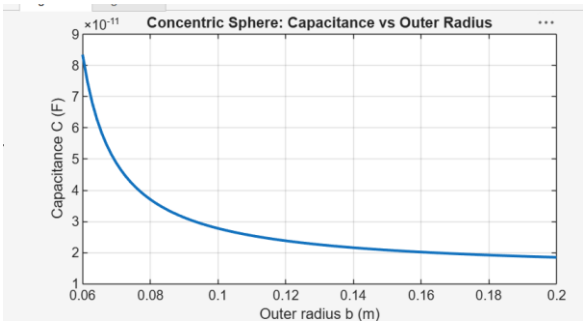
Test Case (iv) – Effect of Outer Radius and Relative Permittivity

PROGRAM:

```
clc; clear;
epsilon0 = 8.854e-12; a = 0.05; er = 2.5;
b = linspace(0.06,0.20,100);
C1 = 4*pi*epsilon0*er.*a.*b./(b-a);
figure; plot(b,C1,'LineWidth',2); grid on;
xlabel('Outer radius b (m)'); ylabel('Capacitance C (F)');
title('Concentric Sphere: Capacitance vs Outer Radius');
er = linspace(1,6,100); b = 0.10;
C2 = 4*pi*epsilon0.*er.*a.*b./(b-a);
figure; plot(er,C2,'LineWidth',2); grid on;
xlabel('Relative permittivity er'); ylabel('Capacitance C (F)');
title('Concentric Sphere: Capacitance vs Relative Permittivity');
```

OUTPUT:





RESULTS:

Thus the program was verified for the capacitance of an isolated sphere and a concentric spherical capacitor for different radii and dielectric constants using MATLAB successfully.

. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained

1	Define spherical geometry and dielectric	Define spherical geometry and dielectric	Correctly identifies isolated sphere radius or inner/outer radii and medium permittivity.	Geometr y is correct with minor omission.	Some dimensions or material properties are unclear.	Uses invalid geometry or material inputs.	25	2.5	
2	Implemen t spherical capacitan ce equations	Implement spherical capacitance equations	Applies the appropriate isolated/con centric sphere formulas correctly.	Formulas are mostly correct with minor errors.	Partial calculations require guidance.	Uses incorrect equations..	25	2.5	
3	Analyze radius and permittivi ty effects	Analyze radius and permittivity effects	Compares cases and validates the expected capacitance trends against theory.	Analysis is mostly correct with minor omission s.	Comparisons or validation are incomplete.	No valid trend analysis is shown.	30	3	
4	Present results and conclusio ns	Present results and conclusions	Reports geometry, units, calculations and conclusions clearly.	Report is clear with minor gaps.	Report lacks some key information..	Results are unclear or missing.	20	2	
Total							100	10	